THE UNIVERSITY OF MELBOURNE

Student number

Semester 2 Assessment, 2022

School of Mathematics and Statistics

MAST30027 Modern Applied Statistics

Reading time: 30 minutes — Writing time: 2 hours — Upload time: 30 minutes

This exam consists of 14 pages (including this page) with 5 questions and 50 total marks

Permitted Materials

- This exam and/or an offline electronic PDF reader, one or more copies of the masked exam template made available earlier, blank loose-leaf paper and a Casio FX-82 calculator.
- One double sided A4 page of notes (handwritten only).
- No headphones or earphones are permitted.

Instructions to Students

- Wave your hand right in front of your webcam if you wish to communicate with the supervisor at any time (before, during or after the exam).
- You must not be out of webcam view at any time without supervisor permission.
- You must not write your answers on an iPad or other electronic device.
- Off-line PDF readers (i) must have the screen visible in Zoom; (ii) must only be used to read exam questions (do not access other software or files); (iii) must be set in flight mode or have both internet and Bluetooth disabled as soon as the exam paper is downloaded.

Writing

- If you are writing answers on the exam or masked exam and you need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

Scanning and Submitting

- **You must not leave Zoom supervision to scan your exam.** Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned exam as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary. Do not leave Zoom supervision until you have confirmed orally with the supervisor that you have received the Gradescope confirmation email.
- **You must not submit or resubmit after having left Zoom supervision.**

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Useful distributions

- Normal(mean = μ, variance = σ^2). The Normal distribution with mean μ and variance σ^2 has the probability density function

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

  and has the expectation $\mathbb{E}(X) = \mu$.

- Inverse-Gaussian(mean = μ, shape = λ). The Inverse-Gaussian with mean = $\mu > 0$ and shape = $\lambda > 0$ has the probability density function

$$f(x; \mu, \lambda) = \sqrt{\frac{\lambda}{2\pi x^3}} e^{-\frac{\lambda(x-\mu)^2}{2\mu^2 x}}$$

  for $x > 0$, and has the expectation $\mathbb{E}(X) = \mu$.

- Exponential(rate = α). The exponential distribution with rate $\alpha > 0$ has the probability density function

$$f(x; \alpha) = \alpha e^{-\alpha x}$$

  for $x \geq 0$, and has the expectation $\mathbb{E}(X) = \frac{1}{\alpha}$.

- Gamma(shape = a, scale = b). The Gamma distribution with shape $a > 0$ and scale $b > 0$ has the probability density function

$$f(x; a, b) = \frac{1}{\Gamma(a) b^a} x^{a-1} e^{-\frac{x}{b}}$$

  for $x > 0$, and has the expectation $\mathbb{E}(X) = ab$.

- Inverse-Gamma(shape = a, scale = b). The Inverse-Gamma distribution with shape $a > 0$ and scale $b > 0$ has the probability density function

$$f(x; a, b) = \frac{b^a}{\Gamma(a)} x^{-a-1} e^{-\frac{b}{x}}$$

  for $x > 0$, and has the expectation $\mathbb{E}(X) = \frac{b}{a-1}$ for $a > 1$.

- Poisson(rate = λ). The Poisson distribution with rate $\lambda > 0$ has the probability mass function

$$f(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

  for $x = 0, 1, \ldots$, and has the expectation $\mathbb{E}(X) = \lambda$.

- Categorical(π_1, π_2). The categorical distribution with event probabilities π_1 and π_2, where $\pi_1 > 0$, $\pi_2 > 0$, and $\pi_1 + \pi_2 = 1$, has the probability mass function

$$f(x; \pi_1, \pi_2) = \pi_x$$

  for $x = 1, 2$, and has the expectation $\mathbb{E}(X) = 2 - \pi_1$.

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Question 1 (7 marks)
Let $X_1, \cdots, X_n$ be independent samples from the exponential distribution with rate $\frac{1}{\lambda}$, i.e.,

$$X_i \sim \text{Exponential}(\frac{1}{\lambda})$$

for $i = 1, \ldots, n$.

Note: the probability density function of the Exponential(α) distribution is given on the second page of the exam.

(a) What is the log-likelihood of λ?

(b) What is the Fisher information?

(c) Find the MLE of λ and its asymptotic distribution.

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Question 2 (7 marks)
The `gala` data in the `faraway` package consists of 30 observations (i.e., 30 Galapagos islands). Here, we consider `Species` as the response and `log(Area)`, `log(Elevation)`, `log(Nearest)`, `log(Scruz+0.1)`, `log(Adjacent)` as predictor variables, and we decide to remove the 25th observation (Santa Cruz island) as an outlier. Examine the R code and output below, and then answer the questions that follow.

```
> library(faraway)
> data(gala)
> modelA <- glm(Species ~ log(Area) + log(Elevation) +
+                   log(Nearest) + log(Scruz+0.1) + log(Adjacent),
+               family=poisson, gala, subset=-25)
> modelB <- glm(Species ~ log(Area) +
+                   log(Nearest) + log(Adjacent),
+               family=poisson, gala, subset=-25)
> summary(modelA)

Call:
glm(formula = Species ~ log(Area) + log(Elevation) + log(Nearest) +
    log(Scruz + 0.1) + log(Adjacent), family = poisson, data = gala,
    subset = -25)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-5.7237  -2.7539  -0.3181   2.6401   7.9333

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)       3.05070    0.30033  10.158  < 2e-16 ***
log(Area)         0.33453    0.01883  17.770  < 2e-16 ***
log(Elevation)    0.05960    0.05743   1.038 0.299325
log(Nearest)     -0.05255    0.01469  -3.578 0.000347 ***
log(Scruz + 0.1)  0.01592    0.02218   0.718 0.472998
log(Adjacent)    -0.08852    0.00696 -12.717  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 2707.88  on 28  degrees of freedom
Residual deviance:  353.42  on 23  degrees of freedom
AIC: 518.32

Number of Fisher Scoring iterations: 5

> summary(modelB)

Call:
glm(formula = Species ~ log(Area) + log(Nearest) + log(Adjacent),
    family = poisson, data = gala, subset = -25)
```

```
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-5.4799  -2.9871  -0.6226   2.5441   7.7881

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    3.384649   0.048760  69.415  < 2e-16 ***
log(Area)      0.352919   0.008598  41.048  < 2e-16 ***
log(Nearest)  -0.047878   0.012378  -3.868  0.00011 ***
log(Adjacent) -0.086616   0.006776 -12.783  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 2707.88  on 28  degrees of freedom
Residual deviance:  354.83  on 25  degrees of freedom
AIC: 515.72

Number of Fisher Scoring iterations: 5

> (phiA <- sum(residuals(modelA, type="pearson")^2/modelA$df.residual))
[1] 16.48824
> (phiB <- sum(residuals(modelB, type="pearson")^2/modelB$df.residual))
[1] 15.27199
> modelC <- glm(Species ~ log(Area) + log(Elevation) +
+             log(Nearest) + log(Scruz+0.1) + log(Adjacent),
+             family=quasipoisson, gala, subset=-25)
> modelD <- glm(Species ~ log(Area)  +
+             log(Nearest) + log(Adjacent),
+             family=quasipoisson, gala, subset=-25)
> anova(modelD, modelC, test="F")
```

(a) Compute the F-statistic from the `anova(modelD, modelC, test="F")` command. What are the degrees of freedom for the F-statistic?

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(b) Calculate the estimate for the coefficient of `log(Area)` and its standard error for `modelD`.

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Question 3 (10 marks)
Let $X_1, \ldots, X_{100}$ be independent random variables where $X_1, \ldots, X_{50}$ follow a mixture of two Normal distributions, and $X_{51}, \ldots, X_{100}$ follow one of the two Normal distributions in the mixture. Specifically, for $i = 1, \ldots, 50$,

$$Z_i \sim \text{Categorical}(p, 1-p),$$

$$X_i | Z_i = 1 \sim \text{Normal}(0, \sigma_1^2) \text{ and } X_i | Z_i = 2 \sim \text{Normal}(0, \sigma_2^2),$$

and for $i = 51, \ldots, 100$,

$$X_i \sim \text{Normal}(0, \sigma_1^2).$$

We aim to obtain the MLE of the parameters $\theta = (p, \sigma_1^2, \sigma_2^2)$ using the EM algorithm.

Note 1: The probability density function of the Normal(μ, σ^2) distribution and the probability mass function of the Categorical(π_1, π_2) distribution are given on the second page of the exam.
Note 2: If your final answers contain $P(Z_i = 1 | X_i, \theta^0)$ or $P(Z_i = 2 | X_i, \theta^0)$, you do not need to provide detailed formulae for them, i.e., your final answers can contain these expressions.

(a) Let $X = (X_1, \ldots, X_{100})$ and $Z = (Z_1, \ldots, Z_{50})$. Derive the expectation of the complete log-likelihood, $Q(\theta, \theta^0) = \mathbb{E}_{Z|X,\theta^0}[\log(P(X, Z|\theta))]$.

(b) Derive the M-step for σ_1^2 and σ_2^2 of the EM algorithm.

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Question 4 (17 marks)
Model: We assume that x and y are independent and follow Normal distributions:

$$x \sim \text{Normal}(\mu, 1^2), \quad y \sim \text{Normal}(0, \frac{1}{\tau}).$$

Prior: We impose the following prior for the μ and τ parameters:

$$p(\mu, \tau) = p(\mu|\tau)p(\tau),$$

$$\mu|\tau \sim \text{Normal}(0, \frac{1}{\tau}), \quad \tau \sim \text{Gamma}(2, 1).$$

Note : The probability density functions of the Normal(μ, σ^2) and the Gamma(a, b) distributions are given on the second page of the exam.

(a) **Gibbs sampling:** we wish to perform posterior inference using the Gibbs sampling. Derive the conditional distribution

$$p(\mu|\tau, x, y).$$

If it is a known distribution, identify the distribution name and its parameters.

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(b) **Metropolis-Hastings algorithm:** we wish to perform posterior inference using the Metropolis-Hastings algorithm. For the current values of the parameters (μ_c, τ_c), we propose new values (μ_n, τ_n) as follows: $\tau_n \sim \text{Exponential}(\frac{1}{\tau_c})$ and $\mu_n \sim \text{Normal}(0, \frac{1}{\tau_n})$. Compute the acceptance probability when $(\mu_c, \tau_c) = (-1, 2), (\mu_n, \tau_n) = (1, 1), x = 0, y = 1$. Show your work.

Note: The probability density functions of the Normal(μ, σ^2) and the Exponential(α) distributions are given on the second page of the exam.

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(c) **Variational Inference (VI):** we wish to apply VI with the mean-field variational family where $q(\mu, \tau) = q_\mu(\mu)q_\tau(\tau)$, and use the CAVI algorithm for optimisation. The CAVI algorithm iteratively optimises each factor as follows while holding the other factors fixed:

$$\begin{aligned} q^*_\mu(\mu) &\propto \exp\{\mathbb{E}_\tau[\log p(\mu, \tau, x, y)]\}, \\ q^*_\tau(\tau) &\propto \exp\{\mathbb{E}_\mu[\log p(\mu, \tau, x, y)]\}, \end{aligned}$$

where the expectations $\mathbb{E}_\tau$ and $\mathbb{E}_\mu$ are taken with respect to $q^*_\tau(\tau)$ and $q^*_\mu(\mu)$, respectively. Derive $q^*_\tau(\tau)$. If it is a known distribution, identify the distribution name and its parameters. The parameters may contain $\mathbb{E}_\mu[\mu]$ or $\mathbb{V}_\mu[\mu]$, which are the mean and variance of $q^*_\mu(\mu)$, respectively.

